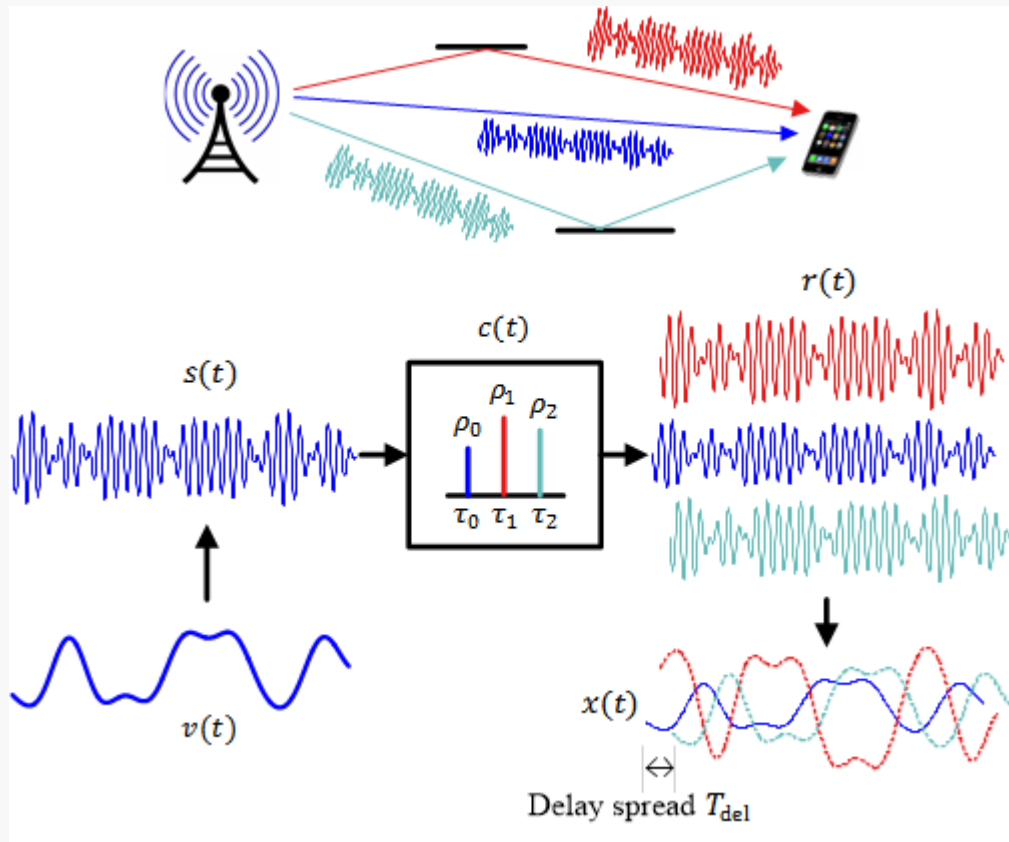


Wireless Channel

Qasim Chaudhari



Wireless Channel

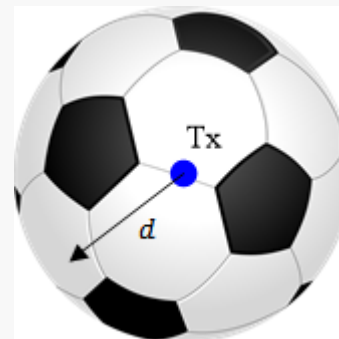
A general case

- **Carrier phase offset:** A channel with a direct path rotated in phase
- **Carrier frequency offset:** A single path with a Doppler shift
- **Symbol timing offset:** A channel with a direct path that arrives with a delay at the Rx
- Solution is the equalizer that acts as a brain of the digital Rx

Large Scale Fading

Power radiates outwards from the Tx in every dimension

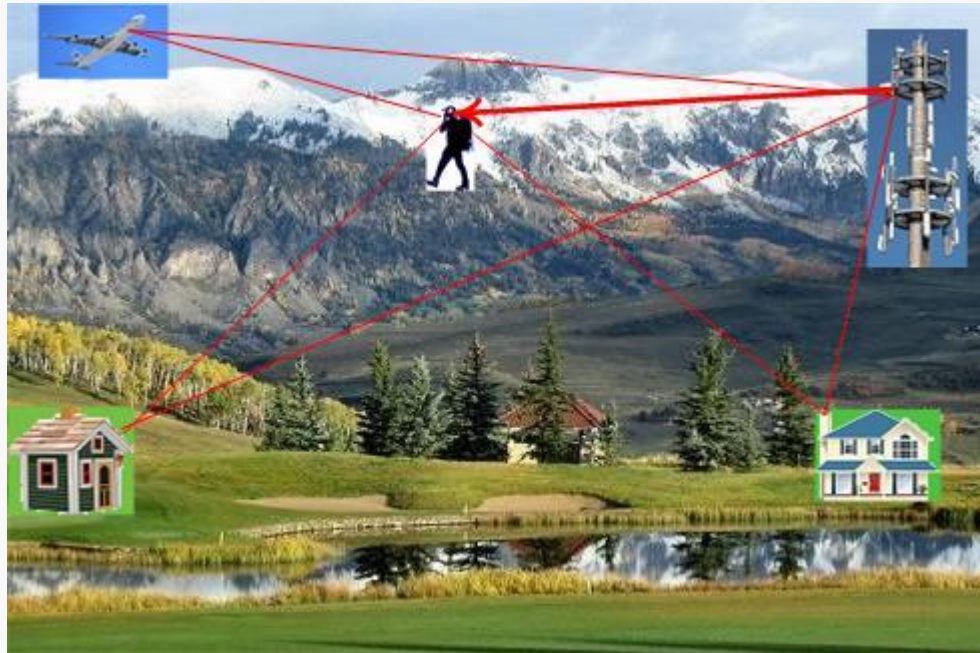
$$\text{Power density} = \frac{P_{Tx}}{4\pi d^2} \quad \text{Watts/m}^2$$



- In free space, power decays with inverse of squared distance
- As a practical rule $P_{Rx} \propto \frac{1}{d^\beta}$ (β is determined by the environment)
- Shadowing

Multipath Distortion

- Assume no carrier wave
- Multiple paths arriving at the Rx with delays of $0.6\ \mu\text{s}$ and $1.1\ \mu\text{s}$



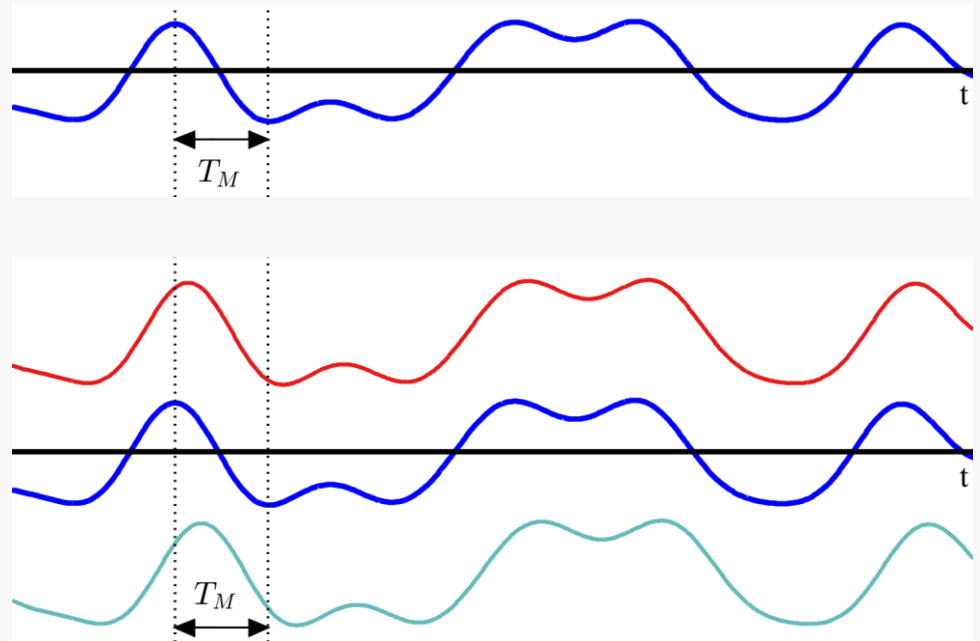
Multipath Distortion

Low data rate scenario

- In early 2000s, our hiker only wants to check his email and/or messages
- Requires a data rate of only $1/T_M = 100$ kbps

$$T_M = \frac{1}{100,000} = 10 \mu\text{s}$$

- Maximum delay of $1.1 \mu\text{s}$ is $0.1T_M$



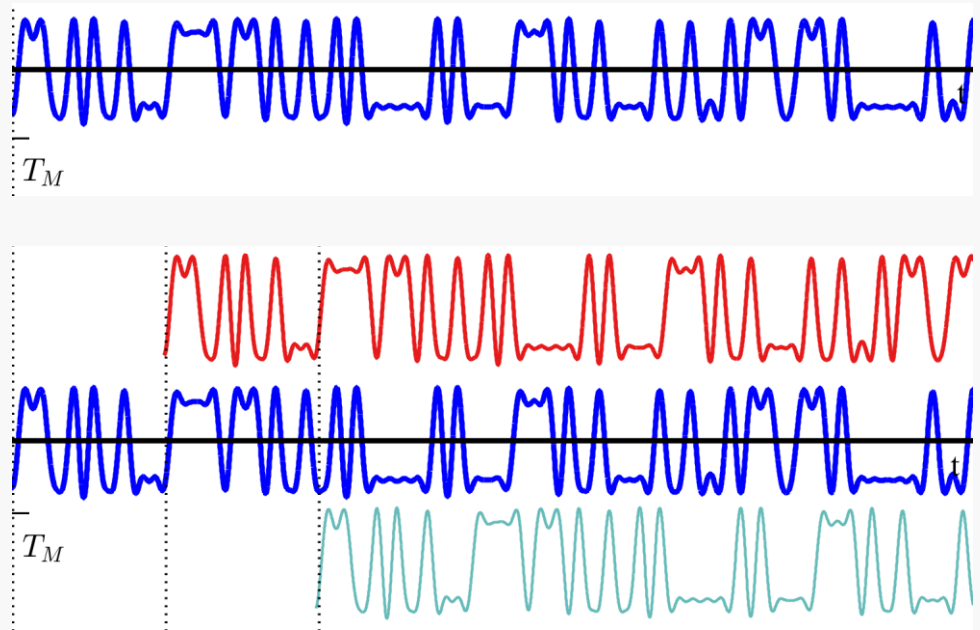
Multipath Distortion

High data rate scenario

- In early 2010s, our hiker wants to sit on top of that peaceful mountain
 - ... to watch YouTube videos
- Data rate of $1/T_M = 10$ Mbps

$$T_M = 0.1 \mu s$$

- Same harmless channel has become harsh for high rate communication!



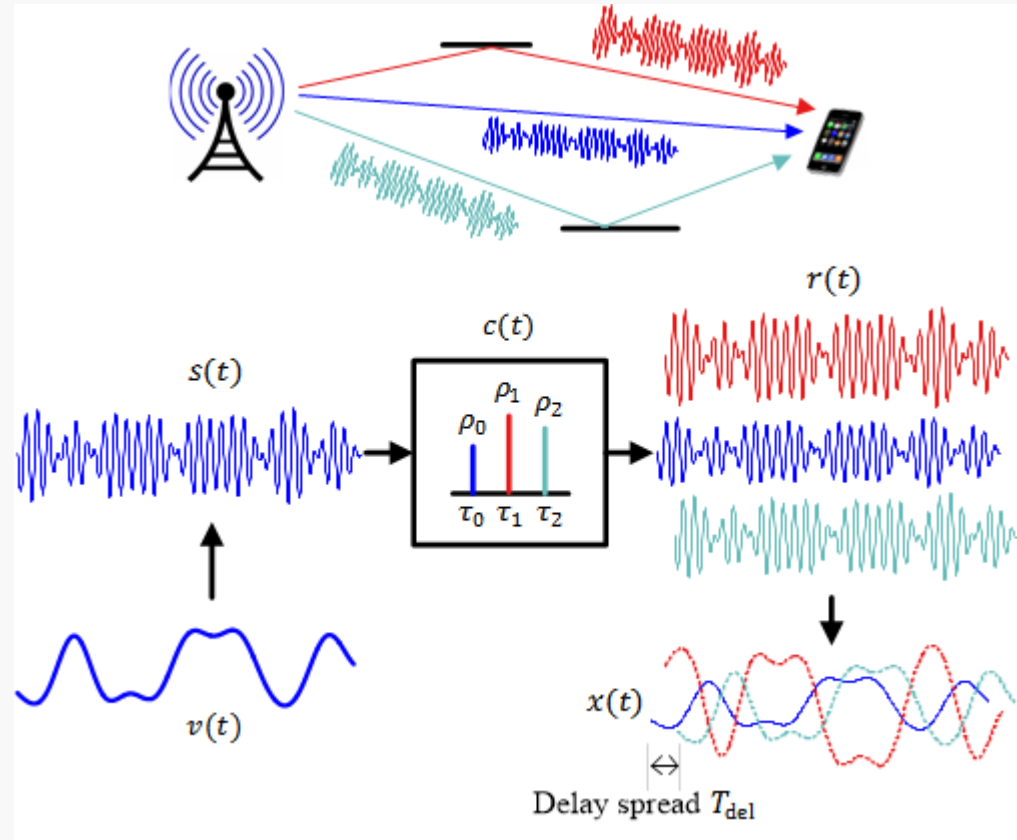
Carrier Wave Enters the Picture

Amplitudes ρ_i :

- Each multipath traverses a different route to the Rx
- Not much change over shorter distances

Delays τ_i :

- Different routes imply different path lengths and hence delays



Carrier Wave Enters the Picture

- Rx waveform

$$r(t) = \sum_{i=0}^{N_{MP}-1} \rho_i \cdot s(t - \tau_i)$$

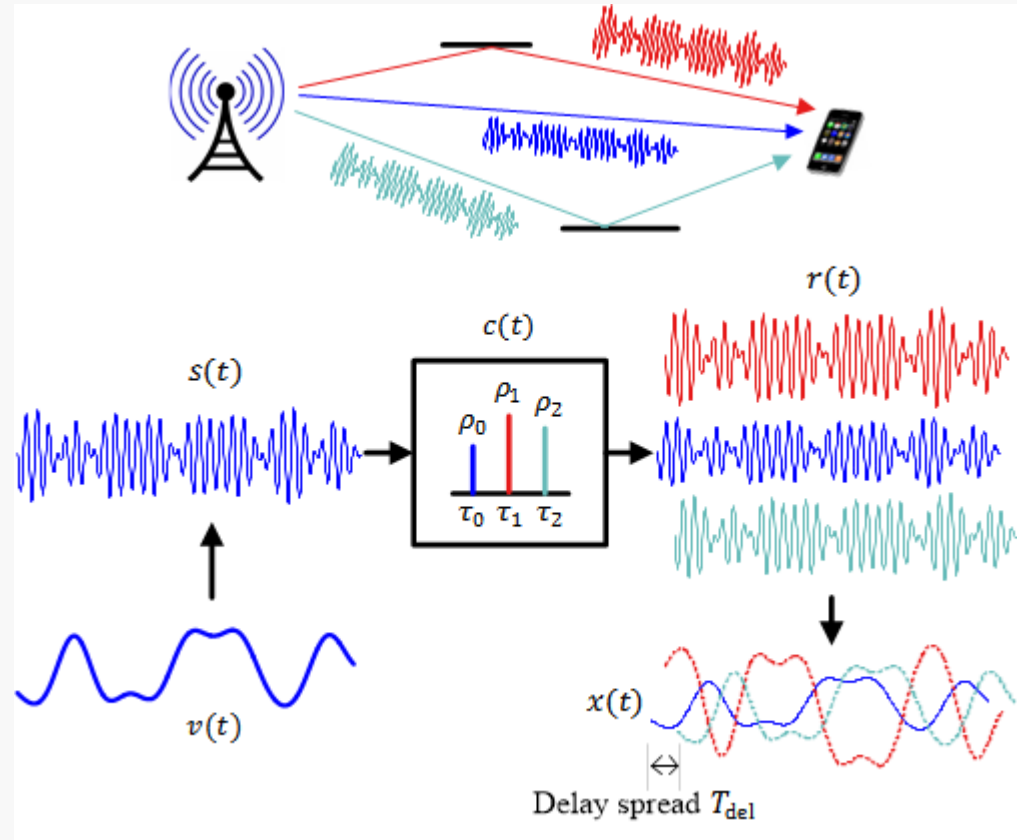
- Baseband channel after downconversion

$$c_B(t) = \sum_{i=0}^{N_{MP}-1} \gamma_i \cdot \delta(t - \tau_i)$$

where γ_i are complex amplitudes!

$$\gamma_{i,I} = \rho_i \cdot \cos 2\pi F_C \tau_i$$

$$\gamma_{i,Q} = -\rho_i \cdot \sin 2\pi F_C \tau_i$$



Delay Spread

Delay of the last significant path

$$T_{\text{Del}} = \tau_{N_{MP}-1} - \tau_0$$

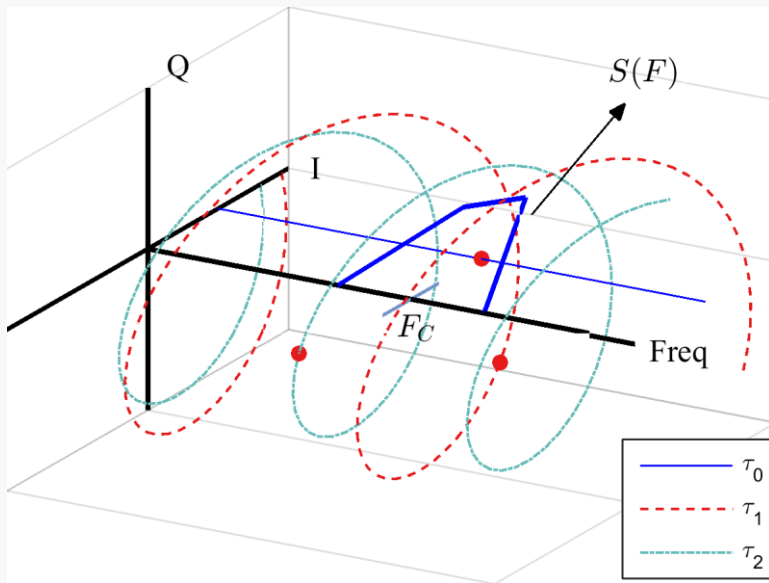
Signifies the channel length

1st out of 4 fundamental quantities through which a wireless channel is categorized

Why Complex Gains?

$$C_I(F) \Big|_{F=F_C} = C_I(F_C) = \sum_{i=0}^{N_{MP}-1} \rho_i \cdot \cos 2\pi F_C \tau_i$$

$$C_Q(F) \Big|_{F=F_C} = C_Q(F_C) = - \sum_{i=0}^{N_{MP}-1} \rho_i \cdot \sin 2\pi F_C \tau_i$$



$$\begin{aligned} s(t) &\longrightarrow S(F) \\ s(t - t_0) &\longrightarrow S(F) \cdot \exp(-j2\pi F t_0) \end{aligned}$$

Example

- Two path channel

$$F_C = 1.9 \text{ GHz}$$

$$\rho_0 = 0.9, \quad \tau_0 = 0 \text{ } \mu\text{s}$$

$$\rho_1 = 0.6, \quad \tau_1 = 5 \text{ } \mu\text{s}$$

$$\gamma_0 = 0.9$$

$$\gamma_{1,I} = \rho_1 \cdot \cos(2\pi F_C \cdot \tau_1) = 0.6$$

$$\gamma_{1,Q} = -\rho_1 \cdot \sin(2\pi F_C \cdot \tau_1) = 0$$

- What happens if τ_0 stays the same but τ_1 is slightly different by 2 ns, i.e., 5.002 μs ?

$$\gamma_{1,I} = 0.1854$$

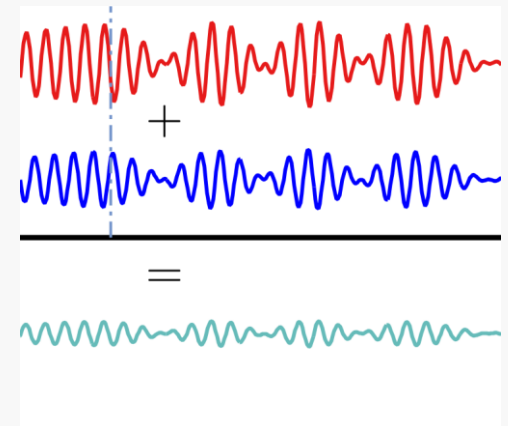
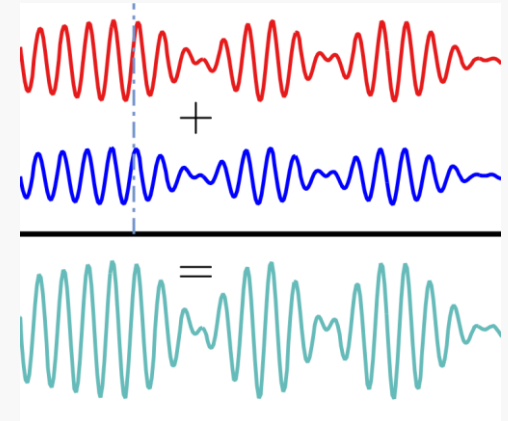
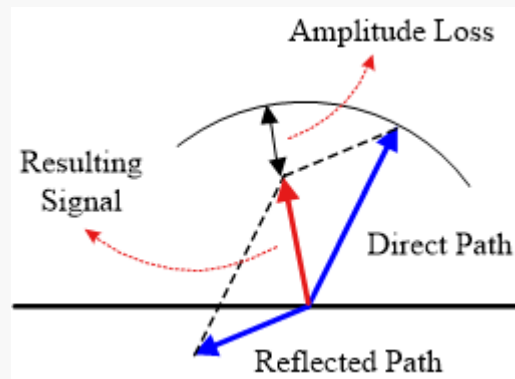
$$\gamma_{1,Q} = 0.5706$$

Delay $\rightarrow$ Phase

Take the term $2\pi F_C \tau_i$

- Phase of each path completes a 2π cycle for each $\tau_i F_C = 1$
- With $F_C = 1.9$ GHz, constructive interference for 15.8 cm
- Destructive interference for 7.9 cm

Vector sum perspective

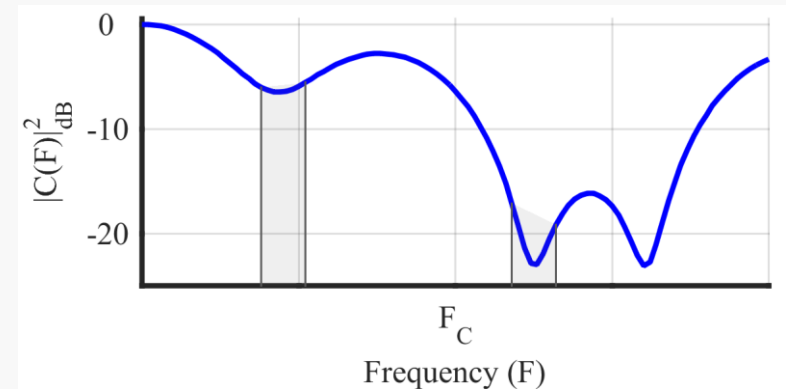


Coherence Bandwidth

Range of frequencies over which the channel frequency response can assumed approximately constant

$$B_C \propto \frac{1}{T_{\text{Del}}}$$

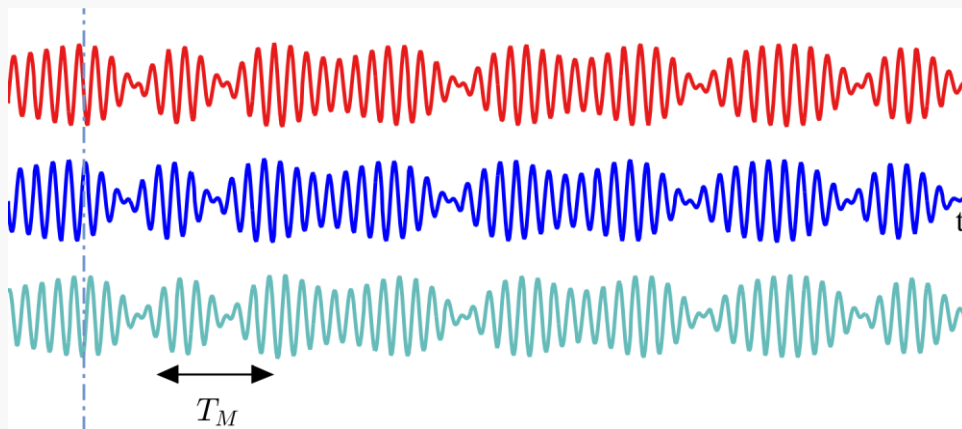
Why?



2nd out of 4 fundamental quantities through which a wireless channel is categorized

Frequency Flat Fading

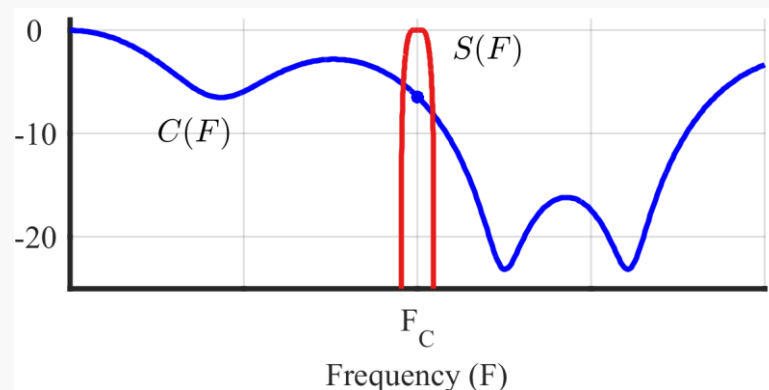
For small delay spread, data symbol values are still (almost) the same



$$T_{\text{Del}} < 0.1T_M$$

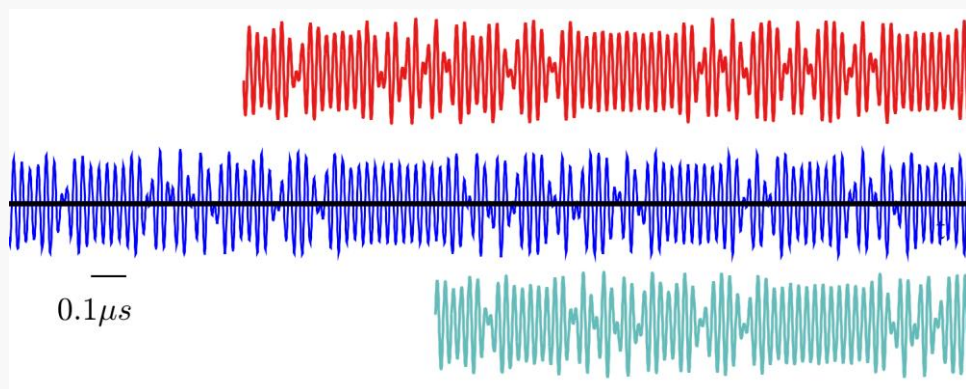


- Channel is just a single impulse
- Signal bandwidth $<$ coherence bandwidth

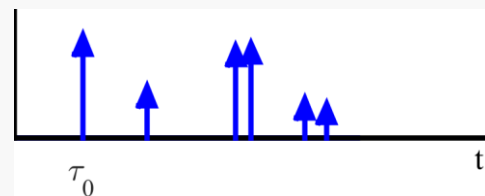


Frequency Selective Fading

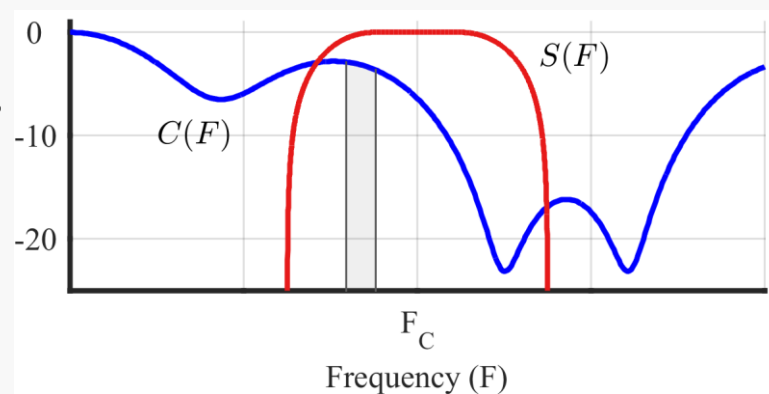
For large delay spread, data symbol values interfere with each other (ISI)



$$T_{\text{Del}} > T_M$$



- Channel response consists of many impulses
- Signal bandwidth $>$ coherence bandwidth

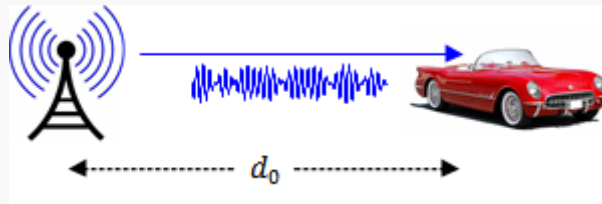


Doppler Shift

T_x , the Rx and the environment are stationary

- A carrier wave $\cos 2\pi F_C t$ impinging on a mobile antenna in a parked car
- Arrives at the Rx τ_0 seconds later

$$\rho_0 \cos 2\pi F_C (t - \tau_0) = \rho_0 \cos 2\pi F_C \left(t - \frac{d_0}{c} \right)$$



Doppler Shift

Now the car is driven at a constant velocity of v m/s opposite to the antenna

- EM wave has to cover a distance of $d_0 + vt$

$$\rho_0 \cos 2\pi F_C \left(t - \frac{d_0 + vt}{c} \right) = \rho_0 \cos 2\pi \left\{ \left(F_C - \frac{v}{c} F_C \right) t - F_C \tau_0 \right\}$$

Doppler shift $F_D = -\frac{v}{c} F_C$

From d_0

- Ratio of velocities between mobile and EM wave
- For a general angle of arrival ϕ

$$F_D = \frac{v \cos \phi}{c} F_C$$

Doppler Shift

A deceptive villain

- Car is moving away ($\phi = \pi$) with a velocity of 60 km/hr at $F_C = 1.9$ GHz,

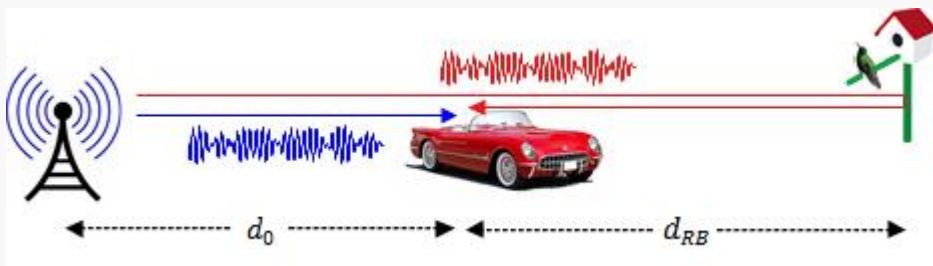
$$F_D = -\frac{16.67}{3 \cdot 10^8} \cdot 1.9 \cdot 10^9 \approx -105 \text{ Hz}$$

- Doppler shift is so small that even a simple PLL would easily track it down!
- As a comparison, assume a ± 20 ppm transceiver crystal

$$\pm \frac{20}{10^6} \times 1.9 \times 10^9 = \pm 38000 \text{ Hz}$$

Doppler Shift

Now there is a multipath copy reflected from the pole of a birdhouse



$$F_D = -\frac{\nu}{c} F_C$$

- Direct path travels a distance equal to $d_0 + \nu t$ meters
- Multipath arrives at the Rx after covering a total of

$$F_D = +\frac{\nu}{c} F_C$$

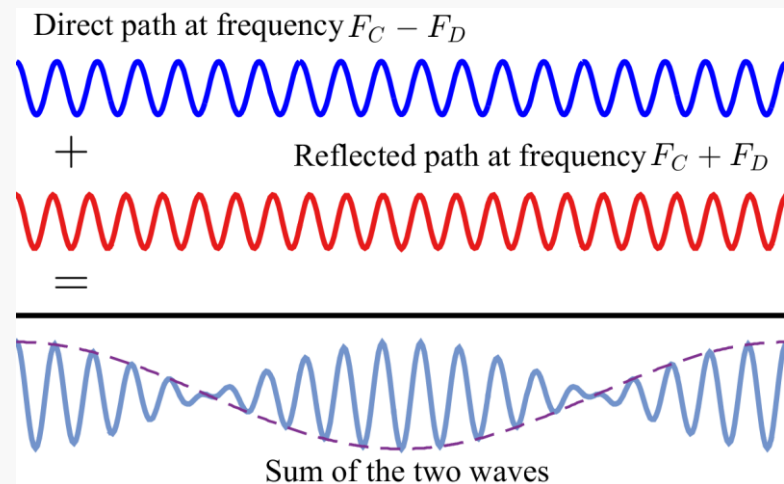
$$\underbrace{d_0 + \nu t + d_{RB} - \nu t}_{\text{Tx to birdhouse}} + \underbrace{d_{RB} - \nu t}_{\text{birdhouse to Rx}} = d_0 + 2d_{RB} - \nu t$$

Doppler Shift

Rx signal is a sum of two sinusoids (sum-product identity)

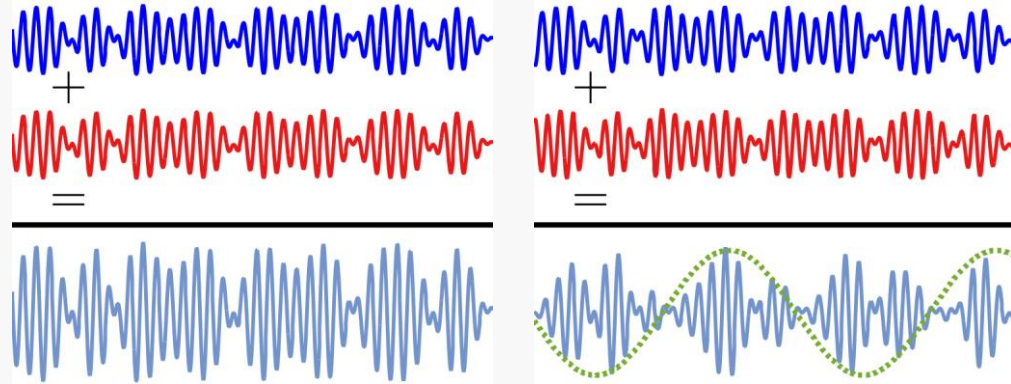
$$r(t) = 2\rho_0 \underbrace{\cos 2\pi (F_C t - F_C \tau_0)}_{\approx \text{GHz}} \underbrace{\cos 2\pi \left(+\frac{\nu}{c} F_C t \right)}_{\approx \text{Hz}}$$

- Low frequency component rides on the high frequency component
 - Creates crests and troughs
 - Beating effect



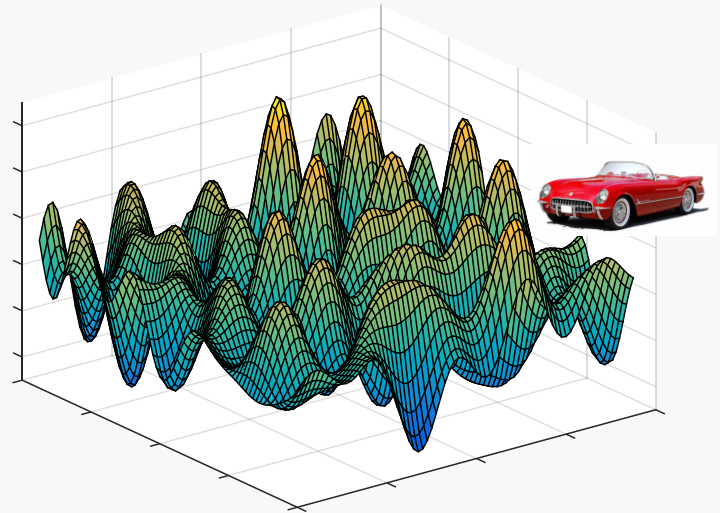
Doppler Shift

1 path: Doppler shift acts on the data like a CFO



2 paths: Envelope with F_D is superimposed on the carrier wave in the form of a product

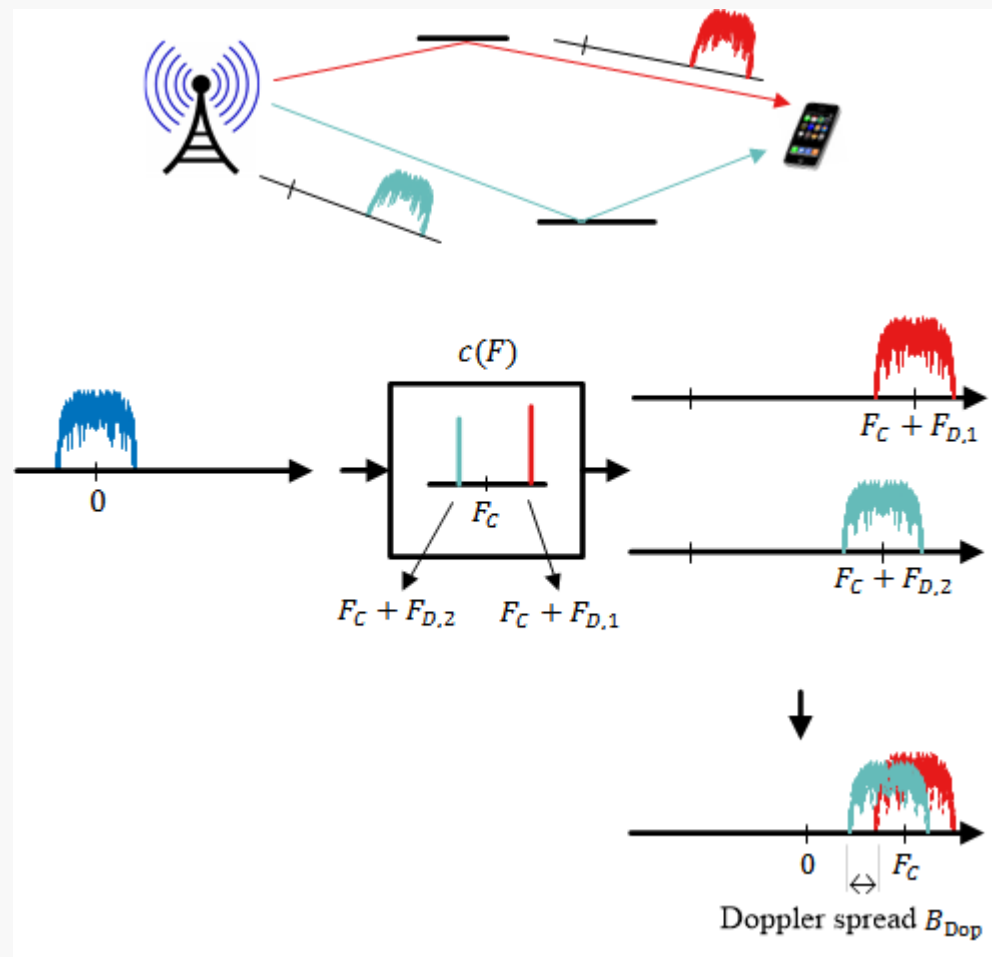
Multiple paths: Random pattern of peaks and valleys



Doppler Spread

Difference between the largest and the smallest Doppler frequency

- Signifies the width of the channel spectrum
- A dual to delay spread
- 3rd out of 4 fundamental quantities for a wireless channel

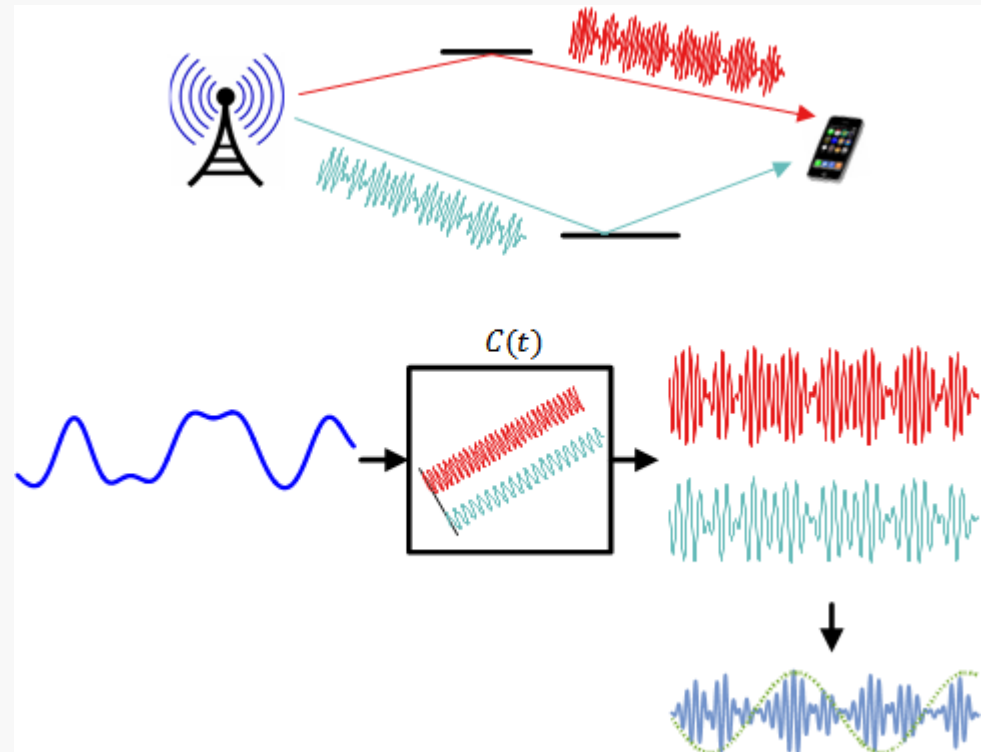


Coherence Time

- Duration over which the time varying amplitude can be assumed approximately constant

$$T_C \propto \frac{1}{B_{\text{Dop}}}$$

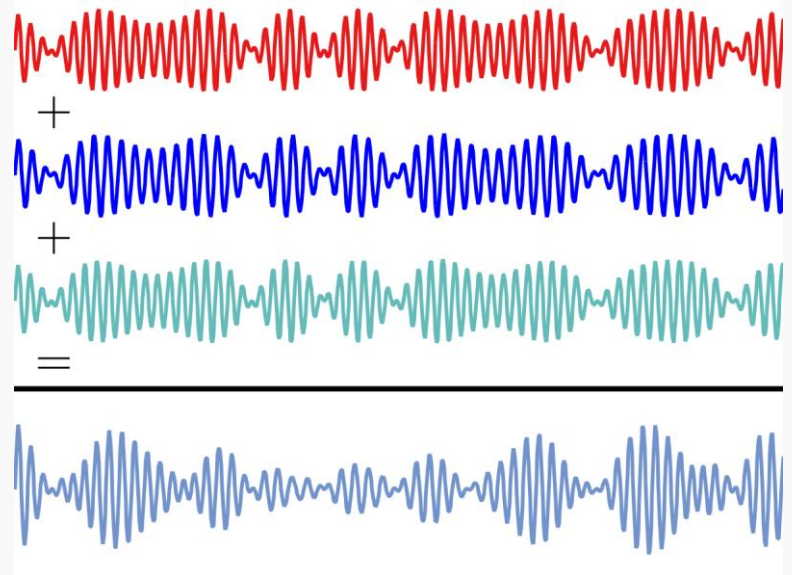
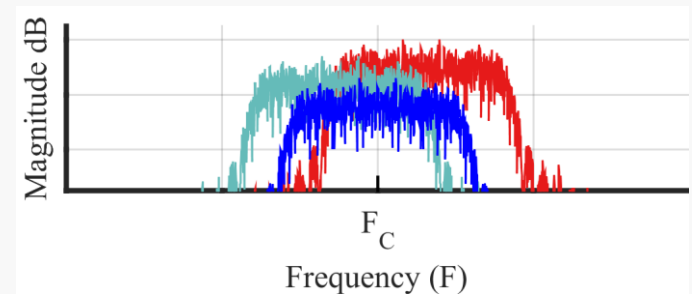
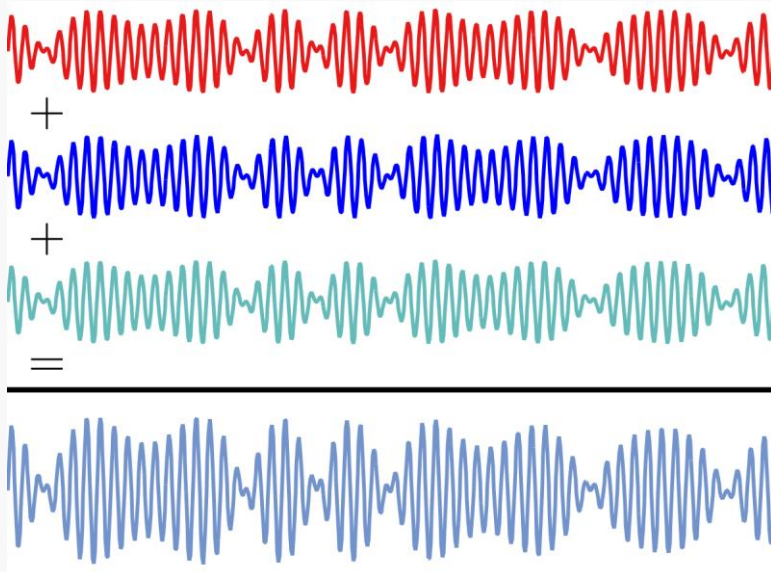
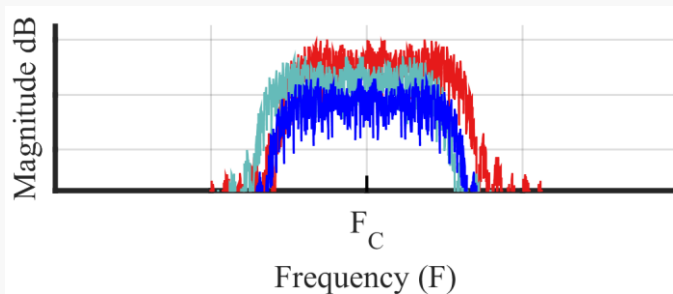
- Similar relation as between T_{Del} and B_C
- 4th out of 4 fundamental quantities through which a wireless channel is categorized



Slow and Fast Fading

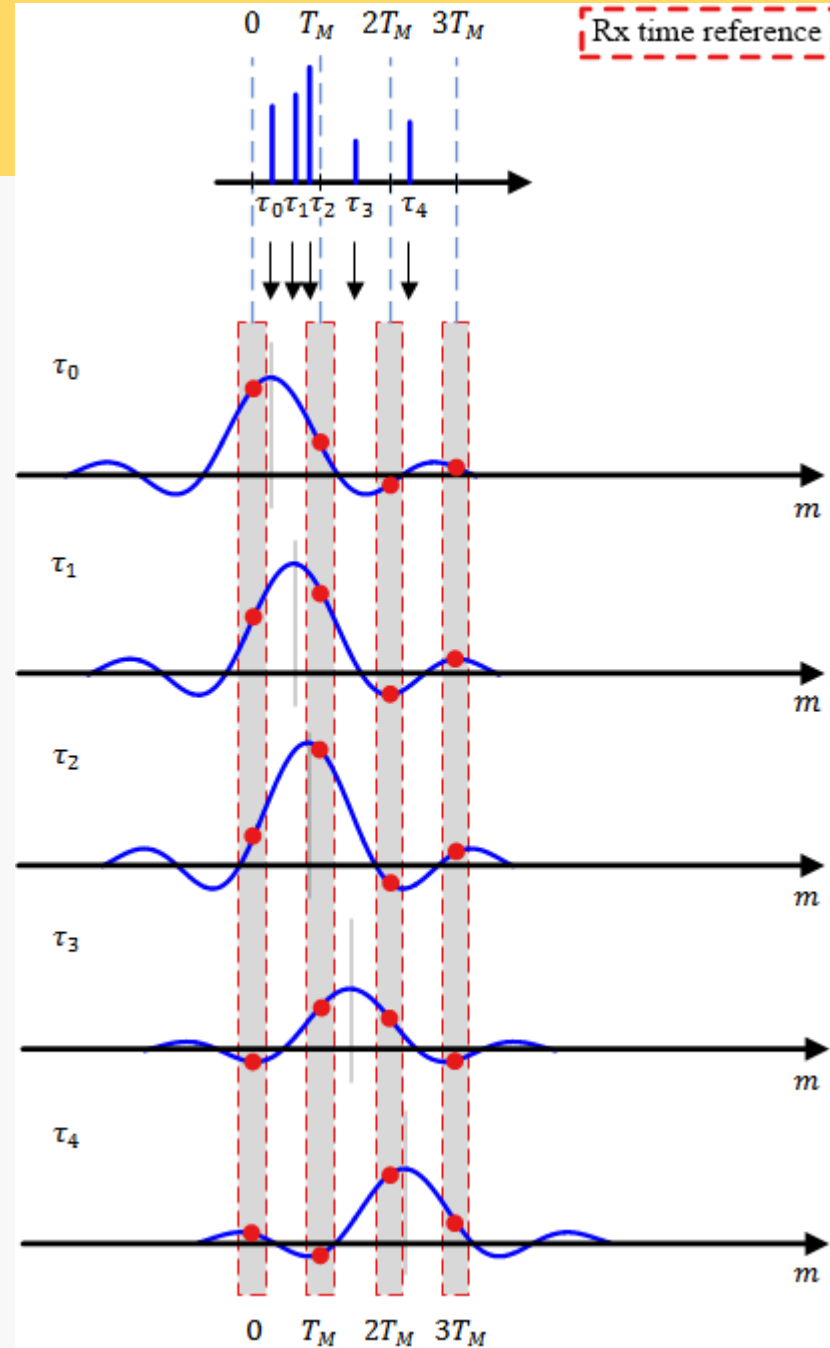
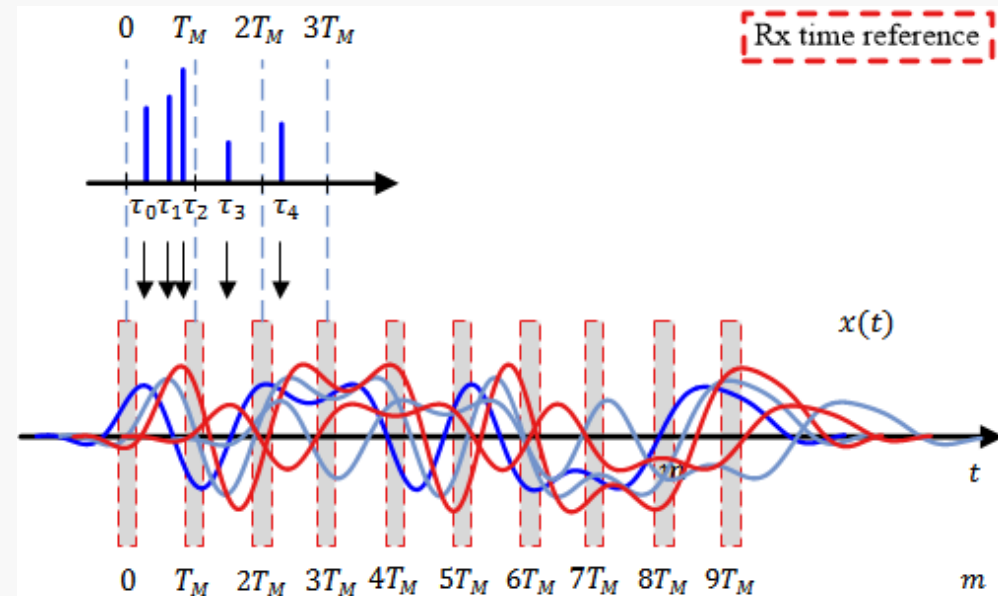
Or time flat and time selective fading

- Coherence time $>$ symbol time
- Coherence time $<$ symbol time



Channel Paths to Taps

- Rx samples the signal at rate $F_S = L/T_M$
- A channel tap is a discrete-time sample that arises from the superposition of all channel paths at that instant



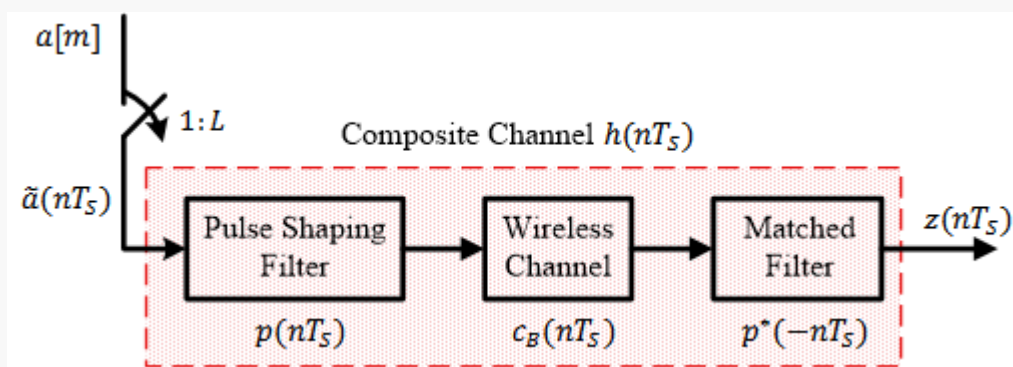
Channel Paths to Taps

- Rx signal is filtered by $p^*(-nT_S)$ which is not a matched filter anymore!

$$z(nT_S) = \tilde{a}(nT_S) * \underbrace{p(nT_S) * c_B(nT_S) * p^*(-nT_S)}_{\text{Composite channel } h(nT_S)}$$

- The final matched filter output is

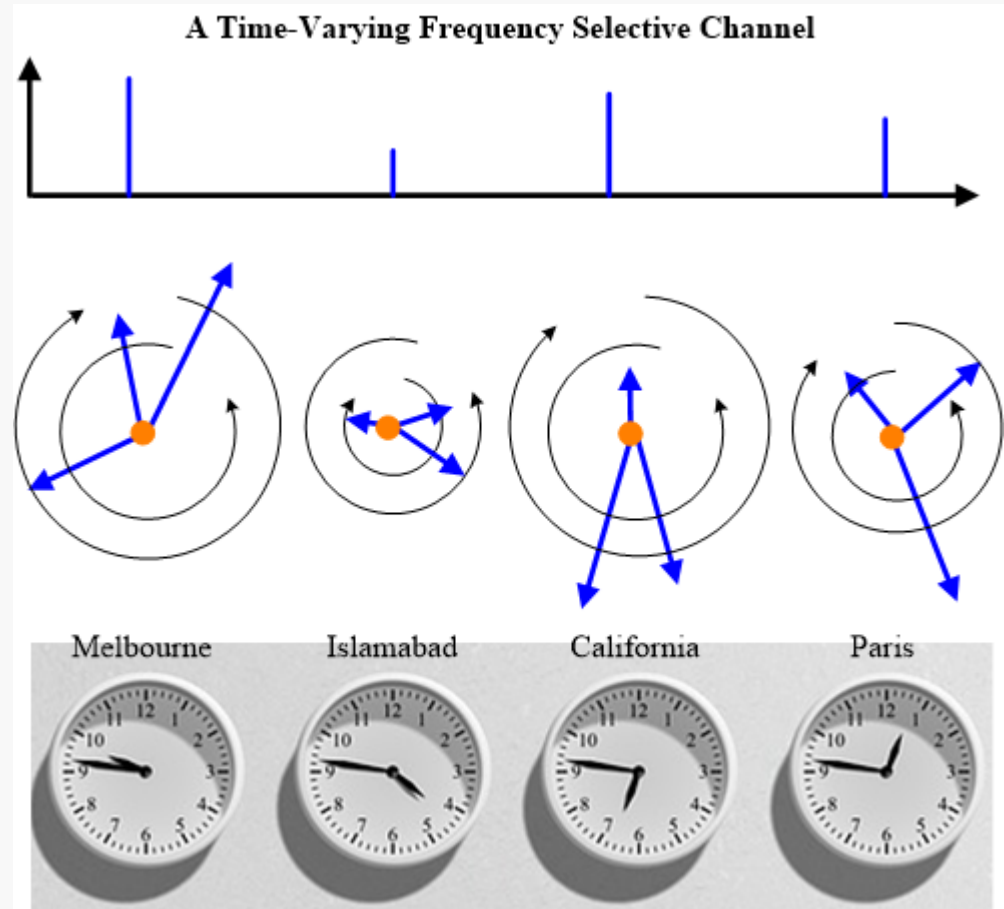
$$z(nT_S) = \sum_m a[m]h(nT_S - mT_M)$$



Time-Varying Channel

Rotating taps in a frequency selective channel

- Quasi-static assumption
- Channel tracking
- Adaptive equalization



Effect of Channel

$$\begin{aligned}\rho_0 &= 1, & \tau_0 &= 0 \\ \rho_1 &= 0.3, & \tau_1 &= T_M\end{aligned}$$

